

Firewall execution in SDN

Check the MAC address against the firewall rules available in the POX controller.

- Is transparent=False, and either Ether type is LLDP or the packet's destination address is a Bridge Filtered Address? If yes, DROP.
- Is the destination multi-cast? If yes, FLOOD.
- Install the flow table entry in the switch so that this entry will be used when a similar packet reappears.

To start this, you will find a skeleton class file at `pox/pox/misc/l2_firewall.py`. This is currently not blocking any traffic and you will need to modify this skeleton code to add your own logic later. Here, I have added some rules to block the packet from a specific address. To test the firewall, put the `l2_firewall.py` in the `pox/pox/misc` directory and run the POX controller as shown below:

```
./pox.py log.level --DEBUG forwarding. l2_firewall
```

The above code runs the firewall on the POX controller and filters the traffic by the rules specified in it.

After the connection between the controller and the switch is established, we can verify the connectivity between all pairs of hosts by typing `pingall` in the Mininet console. Note that when `ping` cannot get through a pair of hosts, you need to wait for the timeout, which takes about 10 seconds.

The execution on an SDN based firewall is shown in Figures 6 and 7.

As shown in Figure 6, Mininet creates a network with three hosts. I have run it on one terminal. In another terminal, the POX controller is executed with the `l2_firewall.py` file, which works as the firewall and filters the traffic as per the rules specified in it.

As shown in Figure 7, I have tested the firewall's rules with the `pingall` command, which runs on one terminal to check the connectivity with all the hosts of the network. The result indicates that one host cannot connect with the other, which can be added as a drop packet rule in the firewall.

You, too, can add rules using the CSV file. So is very easy to modify the rules for packet drops by just changing rules in the CSV files.

In this article, I have tried to demonstrate an SDN based firewall, which has many advantages over the traditional variants. The firewall in SDN is controlled by the central controller rather than the individual device. Also, it is a software based firewall so there's no need for any extra device. The updating rules are also easy to make. 

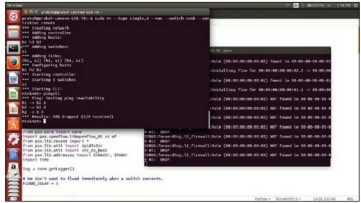


Figure 6: Firewall execution in SDN

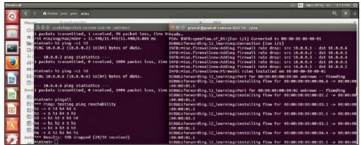


Figure 7: Packet dropped by the firewall in SDN

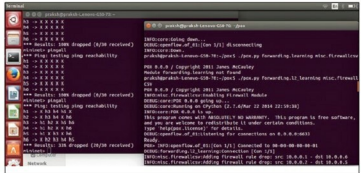


Figure 8: Firewall rules added by CSV file

References

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- [5] <http://opensourcforum.com/2016/01/configuring-an-sdn-controller-in-open-source-mininet-emulator/>
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